“PAPER_{0:D3}” -f [int]# PAPER #76 □ Gamma-Ray Sources: Fermi-LAT + UQFF Emission Model

Title: Fermi-LAT 4FGL Gamma-Ray Source Catalog: UQFF Resonant Mode Emission Predictions

Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
Source Data: QCalc_validation.py (DataSourceURLs: FERMI_LAT, FERMI_4FGL, HEASARC_GAMMA)
Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics,
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2. UQFF Gamma-Ray Emission Model

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Far below any observable threshold ? **Fermi-LAT spectral shapes are unmodified by UQFF.**

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For the Crab Pulsar (PSR J0534+2200, f = 29.65 Hz): - ?_Crab = 2p × 29.65 = 186.3 rad/s - UQFF Resonant: g_R = cos(186.3t) × 10?5 - At emission pulse (t = 0): g_R = 10?5 m/s² maximum - This corresponds to a phase-dependent gravity variation of 3.6×10?8 relative to Newtonian g

The Fermi-LAT 4FGL catalog lists the Crab as **4FGL J0534.5+2201** with flux (100 MeV–100 GeV) = 5.65×10?7 ph/cm²/s. UQFF does not modify the average flux but

predicts a 10?5 amplitude modulation component below 4FGL timing precision for single-pulse analysis but potentially detectable in epoch-folded analysis.

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4. 4FGL Source Comparison Table

4FGL Source	Type	UQFF Prediction	Fermi-LAT Constraint
J0534.5+2201	Pulsar (Crab)	10?5 flux modulation	<1% in phase-folded
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J1103.5+1157 (Mrk 421)	BL Lac	10?5 at ?_jet	Below sensitivity
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Summary

Gamma-Ray Observable	Standard GR+QED	UQFF Prediction	4FGL Status
Average flux	Eddington-based	Unmodified	Match
Spectral shape	Power law/cutoff	Unmodified (m_? ~ 10?³5 eV)	Match
Flux periodicity	Source-dependent	+10?5 modulation	Below sensitivity
Phase-pulse profile	Fixed	+10?5 at peak phase	Not constrained

Source: QCalc_validation.py FERMI_LAT + FERMI_4FGL endpoints | *? = 0.0005/day* | *[SSq] = 0.57* .Groups[1].Value

---

Abstract

The Fermi Gamma-ray Space Telescope 4th Source Catalog (4FGL-DR3, ~3800 sources, 100 MeV1 TeV) provides the definitive census of gamma-ray emitters. The UQFF Resonant mode predicts periodic gamma-ray modulation in blazars and pulsars through the cos(?t) × 10?5 term. For high-energy gamma-ray emission, the UQFF also predicts a vacuum Cherenkov-like correction to the effective photon

mass via the [UA] vacuum density. This paper validates UQFF emission predictions against 4FGL sources using the QCalc_validation.py Fermi endpoints.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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```
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Source: QCalc_validation.py FERMI_LAT + FERMI_4FGL endpoints | ? = 0.0005/day | [SSq] = 0.57

UQFF computed: Canonical UQFF buoyancy parameter U_bi = ?×[SSq]×GM/r^2 = 5.0e-4×0.57×6.67e-11×M/r^2; for solar parameters: U_bi,Sun = 5.7e-4×6.67e-11×1.99e30/$(6.96e8)^2$ = 1.47e+2 m/s^2.

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
Îº	5.0 Ã 10â»â´ dayâ»Â¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
Î²_i	0.60â0.61	Buoyancy coupling coefficient
kâ	1.5	Ug1 DPM-dipole coupling
kâ	1.2	Ug2 outer-bubble charge coupling
kâ	1.8	Ug3 string-rotation coupling
kâ	2.0	Ug4 vacuum-concentration coupling
Î·	10â»Â²Â²	Inertia tensor scale
E_react(0)	10â´â¶ J	Reference reactive energy

A.2 F_U Master Equation (Complete â 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\mathrm{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`

Term	Description	Implementation
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
â□□Î£Î»áµ¢Â·Uáµ¢Â·F_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
Î»â□□=10â□»Â¹â□°, Î»â□□=10â□»Â¹Â², Î»â□□=10â□»Â¹Â¹, Î»â□□=10â□»Â¹Â³ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q\cos(\Delta\omega\, t)igr)$$

Symbol	Value	Description
Ï□_c	10Â¹â□µ kg/mÂ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Î□Ï□	2Ï□/(434Â·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, â□¦)	Multi-scale field interactions
Buoyant	Î²_i Ã□ Ubi	Expanding nebulae, stellar winds
Superconductive	Um Ã□ (1+10Â¹Â³Â·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.